Quadratics & Polynomials

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Factoring quadratics

Solve $x^2 + 7x + 12 = 0$ by factoring.

$(x + 3)(x + 4) = 0$, so $x = -3$ or $x = -4$.

2 [Foundational] • Target: Vertex form and completing the square

Write $y = x^2 - 6x + 5$ in vertex form by completing the square, and state the vertex.

$x^2 - 6x + 5 = (x - 3)^2 - 9 + 5 = (x - 3)^2 - 4$. Vertex: $(3, -4)$.

3 [Foundational] • Target: Vertex form and completing the square

Solve $x^2 + 8x + 3 = 0$ by completing the square.

$x^2 + 8x = -3$, so $(x + 4)^2 - 16 = -3$, giving $(x + 4)^2 = 13$, so $x = -4 \pm \sqrt{13}$.

4 [Foundational] • Target: The quadratic formula and discriminant

Solve $3x^2 - 5x - 2 = 0$ using the quadratic formula.

$a = 3, b = -5, c = -2$. Discriminant $= 25 + 24 = 49$. $x = \frac{5 \pm 7}{6}$, so $x = 2$ or $x = -\frac{1}{3}$.

5 [Foundational] • Target: The quadratic formula and discriminant

Find the discriminant of $2x^2 + 3x + 5 = 0$ and state the nature of its roots.

Discriminant $= 3^2 - 4(2)(5) = 9 - 40 = -31$, which is negative, so the equation has no real roots.

6 [Foundational] • Target: The quadratic formula and discriminant

For what values of k does $x^2 + kx + 16 = 0$ have exactly one repeated real root?

A repeated root occurs when the discriminant is zero: $k^2 - 4(1)(16) = 0$, so $k^2 = 64$, giving $k = \pm 8$.

7 [Foundational] • Target: Graphing parabolas

The graph of $y = x^2 - 2x - 3$ is shown below. Use the graph to state the x-intercepts, then confirm your answer algebraically by factoring.

From the graph, the parabola crosses the x-axis at $x = -1$ and $x = 3$. Algebraically:
$x^2 - 2x - 3 = (x - 3)(x + 1) = 0$, so $x = 3$ or $x = -1$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Graphing parabolas

The graph of $y = -2x^2 + 8x - 3$ opens which direction, and what is the x-coordinate of its vertex?

Since $a = -2 < 0$, the parabola opens downward. Vertex: $x = -\frac{b}{2a} = -\frac{8}{2(-2)} = 2$.

9 [Intermediate] • Target: Polynomial division and the remainder theorem

Use synthetic division to divide $x^3 - 2x^2 - 5x + 6$ by $(x - 1)$, and state the quotient and remainder.

Using synthetic division with root 1 on coefficients $1, -2, -5, 6$: bring down 1; $1(1) = 1$, $-2 + 1 = -1$; $-1(1) = -1$, $-5 + (-1) = -6$; $-6(1) = -6$, $6 + (-6) = 0$. Quotient: $x^2 - x - 6$, remainder: 0.

10 [Intermediate] • Target: Polynomial division and the remainder theorem

Use the Remainder Theorem to find the remainder when $x^3 + 4x^2 - x + 2$ is divided by $(x + 2)$.

By the Remainder Theorem, the remainder equals $f(-2)$: $(-2)^3 + 4(-2)^2 - (-2) + 2 = -8 + 16 + 2 + 2 = 12$.

11 [Intermediate] • Target: Quadratic word problems

A ball is thrown upward so that its height is modeled by $h(t) = -16t^2 + 64t + 5$ (in feet), where t is time in seconds. Find the time at which the ball reaches its maximum height, and find that maximum height.

Maximum occurs at $t = -\frac{b}{2a} = -\frac{64}{2(-16)} = 2$ seconds. $h(2) = -16(4) + 64(2) + 5 = -64 + 128 + 5 = 69$ feet.

12 [Intermediate] • Target: Quadratic word problems

The area of a rectangle is $x^2 + 2x - 15$ square meters, and its width is $x - 3$. Find its length in terms of x by factoring.

$x^2 + 2x - 15 = (x + 5)(x - 3)$. Since width $= x - 3$, the length is $x + 5$.

13 [Intermediate] • Target: Sum and product of roots

For $x^2 - 7x + 10 = 0$, find the sum and product of its roots without solving, using $\text{sum} = -\frac{b}{a}$ and $\text{product} = \frac{c}{a}$.

Sum $= -\frac{-7}{1} = 7$. Product $= \frac{10}{1} = 10$.

14 [Intermediate] • Target: Sum and product of roots

A quadratic equation has roots 3 and -5. Write the equation in standard form.

$(x - 3)(x + 5) = x^2 + 2x - 15 = 0$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: A well-designed word problem: maximizing weekly profit** • **Cross-Lesson: 3.01 Functions & Graphs**

This question has three parts. A company's weekly profit (in thousands of dollars) from producing x hundred units is modeled by $P(x) = -2x^2 + 20x - 32$. (a) Factor $P(x)$ to find the break-even production levels, where profit is zero. (b) Find the vertex of the parabola to determine the production level that maximizes profit, and state the maximum profit. (c) State the range of production levels (in hundreds of units) for which profit is positive.

(a) $P(x) = -2(x^2 - 10x + 16) = -2(x - 2)(x - 8)$. Break-even at $x = 2$ and $x = 8$ (hundreds of units). (b) Vertex: $x = -\frac{b}{2a} = -\frac{20}{2(-2)} = 5$. $P(5) = -2(25) + 20(5) - 32 = -50 + 100 - 32 = 18$, so the maximum profit is 18 thousand dollars at $x = 5$ hundred units (500 units). (c) Profit is positive between the roots: $2 < x < 8$, i.e. between 200 and 800 units.

16 **[Challenging]** • **Target: Quadratic word problems** • **Cross-Lesson: 4.01 Ratios, Rates & Data**

Convert the quadratic function $y = 2x^2 - 12x + 11$ into vertex form $y = a(x - h)^2 + k$. State the coordinates of the vertex and the line of symmetry.

Step 1: Factor the leading coefficient $a = 2$ from the x-terms: $y = 2(x^2 - 6x) + 11$ Step 2: Complete the square inside parentheses. Half of -6 is -3, and $(-3)^2 = 9$:
$y = 2(x^2 - 6x + 9 - 9) + 11 = 2(x - 3)^2 - 2(9) + 11 = 2(x - 3)^2 - 18 + 11$ Step 3: Simplify to vertex form: $y = \mathbf{2(x - 3)^2 - 7}$ The vertex is $\mathbf{(3, -7)}$ and the axis of symmetry is $\mathbf{x = 3}$.

17 **[Challenging]** • **Target: Sum and product of roots** • **Cross-Lesson: 5.01 Statistics & Probability**

A parabola has its vertex at $(3, -4)$ and passes through the point $(1, 0)$. (a) Write the equation of the parabola in standard form $y = ax^2 + bx + c$. (b) Determine the coordinates of the other x-intercept.

(a) Use vertex form: With vertex $(h, k) = (3, -4)$: $y = a(x - 3)^2 - 4$ Substitute $(x, y) = (1, 0)$ to find a: $0 = a(1 - 3)^2 - 4 \Rightarrow 0 = 4a - 4 \Rightarrow 4a = 4 \Rightarrow a = 1$. Expand to standard form:
$y = 1(x^2 - 6x + 9) - 4 = \mathbf{x^2 - 6x + 5}$. (b) Find the other x-intercept: By symmetry, the x-intercepts are equidistant from the axis of symmetry $x = 3$. Since one intercept is at $x = 1$ (distance $3 - 1 = 2$), the other intercept is at $x = 3 + 2 = 5$, giving $\mathbf{(5, 0)}$. (V.

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18 [Challenging] • Target: A well-designed word problem: maximizing weekly profit • Cross-Lesson: 6.01 Geometry

If $x - 2$ is a factor of the polynomial $P(x) = x^3 - kx^2 + 3x - 10$, find the value of k, and determine all real roots of $P(x)$.

Step 1: By the Factor Theorem, $x - 2$ is a factor $\Leftrightarrow P(2) = 0$:

$P(2) = (2)^3 - k(2)^2 + 3(2) - 10 = 0 \Rightarrow 8 - 4k + 6 - 10 = 0 \Rightarrow 4 - 4k = 0 \Rightarrow \mathbf{k = 1}$. Step 2: Substitute $k = 1$ into $P(x) = x^3 - x^2 + 3x - 10$ and divide by $(x - 2)$ using polynomial long division or synthetic division:

$x^3 - x^2 + 3x - 10 = (x - 2)(x^2 + x + 5)$. Step 3: Solve $x^2 + x + 5 = 0$. Its discriminant is \(\Delta = (1)^2 - 4(1)(5) = 1 - 20 = -19

19 [Challenging] • Target: Factoring quadratics • Cross-Lesson: 7.01 Trigonometry & Additional Topics

Solve the quadratic equation $3x^2 + 10x - 8 = 0$ by factoring.

Step 1: Find two integers whose product is $a \cdot c = 3(-8) = -24$ and whose sum is $b = 10$. The factors are $+12$ and -2. Step 2: Split the middle term and factor by grouping:

$3x^2 + 12x - 2x - 8 = 0 \Rightarrow 3x(x + 4) - 2(x + 4) = 0 \Rightarrow (3x - 2)(x + 4) = 0$ Step 3: Set each factor to zero:

$3x - 2 = 0 \Rightarrow x = \frac{2}{3}, \quad x + 4 = 0 \Rightarrow x = -4$ The solutions are $\mathbf{x = \frac{2}{3}}$ and $\mathbf{x = -4}$.

20 [Challenging] • Target: Vertex form and completing the square • Cross-Lesson: 8.01 Exponentials & Logarithms

Use the quadratic formula to solve $x^2 + 2x - 8 = 0$.

Step 1: Identify coefficients: $a = 1$, $b = 2$, $c = -8$. Step 2: Apply the quadratic formula:

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-2 \pm \sqrt{2^2 - 4(1)(-8)}}{2(1)} = \frac{-2 \pm \sqrt{4 + 32}}{2} = \frac{-2 \pm \sqrt{36}}{2}$ Step 3: Evaluate roots:

$x = \frac{-2 \pm 6}{2} \Rightarrow x_1 = \frac{4}{2} = \mathbf{2}, \quad x_2 = \frac{-8}{2} = \mathbf{-4}$ The solutions are $\mathbf{x = 2}$ and $\mathbf{x = -4}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Factoring quadratics

Find the vertex of the parabola $y = -2x^2 - 28x - 92$.

a. $(-7, 6)$ ✓

b. $(-5, 6)$

c. $(7, 6)$

d. $(-7, -6)$

$h = -\frac{b}{2a} = -\frac{-28}{-4} = -7$. $k = y(-7) = -2(-7)^2 + 196 - 92 = 6$. Vertex: $(-7, 6)$.

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22 [Foundational] • Target: Vertex form and completing the square

What is the sum of the roots of the quadratic equation $3x^2 - 11x + 2 = 0$?

a. $-\frac{2}{3}$

b. $-\frac{11}{3}$

c. $\frac{11}{3}$ ✓

d. $\frac{2}{3}$

For any quadratic equation $ax^2 + bx + c = 0$, Vieta's formulas state that: - Sum of roots: $r_1 + r_2 = -\frac{b}{a}$ - Product of roots: $r_1 r_2 = \frac{c}{a}$ Here, $a = 3$, $b = -11$, and $c = 2$. Thus, the sum of the roots is: $r_1 + r_2 = -\frac{-11}{3} = \frac{11}{3}$.

23 [Foundational] • Target: The quadratic formula and discriminant

The graph of the quadratic function $y = x^2 - 6x + 8$ is shown below. What are the coordinates of the vertex of the parabola?

a. **(3, -1)** ✓

b. (3, 1)

c. (1, 3)

d. (-1, 3)

The vertex form of a quadratic equation is $y = a(x - h)^2 + k$, where (h, k) is the vertex. We can convert the given equation $y = x^2 - 6x + 8$ to vertex form by completing the square. First, group the x terms: $y = (x^2 - 6x) + 8$. To complete the square for $x^2 - 6x$, take half of the coefficient of x (-6), square it ((-3)^2 = 9), and add and subtract it inside the parentheses: $y = (x^2 - 6x + 9 - 9) + 8$. Rearrange to form the perfect square trinomial: $y = (x^2 - 6x + 9) - 9 + 8$. Simplify.

24 [Foundational] • Target: Graphing parabolas

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown below. The parabola has its vertex at $(-2, 8)$ and passes through the point $(0, 4)$. Determine the value of c.

a. 4 ✓

b. -2

c. 8

d. 2

The general form of a quadratic function is $f(x) = ax^2 + bx + c$. The graph of this function is a parabola. The vertex of the parabola is given as $(-2, 8)$. A key property of a parabola is that the y-intercept occurs where $x = 0$. We are given that the parabola passes through the point $(0, 4)$. This means that when $x = 0$, $f(x) = 4$. Substituting $x = 0$ into the general form of the quadratic function, we get: $f(0) = a(0)^2 + b(0) + c$ $f(0) = 0 + 0 + c$ $f(0) = c$ Since we know that $f(0) = 4$, we can conclude t.

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25 [Foundational] • Target: Polynomial multiplication and special products

The graph of the quadratic function $y = x^2 - 6x + 9$ is a parabola. What is the x-coordinate of the vertex of this parabola?

a. 3 ✓

b. -3

c. 0

d. 6

The standard form of a quadratic function is $y = ax^2 + bx + c$. The x-coordinate of the vertex of a parabola given in this form is $x = -\frac{b}{2a}$. In the given function, $y = x^2 - 6x + 9$, we have $a = 1$ and $b = -6$. Therefore, the x-coordinate of the vertex is $x = -\frac{-6}{2(1)} = -\frac{-6}{2} = 3$.

26 [Foundational] • Target: Polynomial division and the remainder theorem

When polynomial $P(x) = 2x^3 - 3x^2 + 5x - 7$ is divided by $(x - 1)$, what is the remainder?

a. -3 ✓

b. 3

c. -7

d. 0

By the Remainder Theorem, remainder = $P(1) = 2 - 3 + 5 - 7 = -3$.

27 [Foundational] • Target: Quadratic word problems

For $2x^2 + 14x + 9 = 0$, what is the sum of the roots?

a. 7

b. -7 ✓

c. -6

d. 9/2

sum of roots $= -\frac{b}{a} = -\frac{14}{2} = -7$

28 [Foundational] • Target: Sum and product of roots

What is the product of the solutions to the quadratic equation $3x^2 + 11x - 12 = 0$?

a. -4 ✓

b. 4

c. $-\frac{11}{3}$

d. -12

By Vieta's formulas, product of roots = $c/a = -12/3 = -4$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: A well-designed word problem: maximizing weekly profit

What is the sum of the solutions to the quadratic equation $2x^2 - 8x + 5 = 0$?

a. 4 ✓

b. -4

c. $\frac{5}{2}$

d. 8

By Vieta's formulas, sum of roots = $-b/a = -(-8)/2 = 4$.

30 [Intermediate] • Target: Factoring quadratics

The graph of the quadratic function $y = ax^2 + bx + c$ is shown below. The parabola intersects the x-axis at points $(1, 0)$ and $(p, 0)$, and has a y-intercept of $(0, -3)$. If the axis of symmetry of the parabola is the line $x = 2$, find the value of p.

a. **3** ✓

b. 4

c. 5

d. 6

The roots of the quadratic equation $ax^2 + bx + c = 0$ are the x-intercepts of the parabola. We are given one root as $x_1 = 1$. The axis of symmetry of a parabola given by $y = ax^2 + bx + c$ is $x = -\frac{b}{2a}$. We are given that the axis of symmetry is $x = 2$. For a quadratic function, the axis of symmetry is located exactly halfway between the roots. If the roots are x_1 and x_2, then the axis of symmetry is $x = \frac{x_1 + x_2}{2}$. We are given $x_1 = 1$ and the axis of symmetry $x = 2$. Let the oth.

31 [Intermediate] • Target: Vertex form and completing the square

What is the vertex of the parabola $y = x^2 - 6x + 11$?

a. $(3, 2)$ ✓

b. $(-3, 2)$

c. $(3, -2)$

d. $(6, 11)$

Applying Quadratics & Polynomials principles, $h = 3$, $k = 9 - 18 + 11 = 2$. Therefore, the verified correct option is $(3, 2)$.

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32 [Intermediate] • Target: The quadratic formula and discriminant

What are the solutions to the quadratic equation $x^2 - 10x + 20 = 0$?

a. $10 \pm \sqrt{5}$

b. 5 ± 5

c. $-5 \pm \sqrt{5}$

d. $5 \pm \sqrt{5}$ ✓

Apply the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ where $a = 1$, $b = -10$, and $c = 20$:
$x = \frac{-(-10) \pm \sqrt{(-10)^2 - 4(1)(20)}}{2(1)}$ $x = \frac{10 \pm \sqrt{100 - 80}}{2} = \frac{10 \pm \sqrt{20}}{2}$ $x = \frac{10 \pm 2\sqrt{5}}{2} = 5 \pm \sqrt{5}$.

33 [Intermediate] • Target: Graphing parabolas

If $(x - 3)$ is a factor of the polynomial $P(x) = 2x^3 - kx^2 + 1x - 3$, what is the value of the constant k?

a. 6 ✓

b. 3

c. 8

d. -6

By the Factor Theorem, if $(x - 3)$ is a factor of $P(x)$, then $P(3) = 0$. Substitute $x = 3$ into $P(x)$:
$P(3) = 2(3)^3 - k(3)^2 + 1(3) - 3 = 0$ $54 - 9k + 3 - 3 = 0$ $54 = 9k$ Divide by 9: $k = 6$.

34 [Intermediate] • Target: Polynomial multiplication and special products

The quadratic function $f(x) = 2x^2 - 12x + 23$ is rewritten in vertex form $f(x) = a(x - h)^2 + k$. What is the minimum value of $f(x)$?

a. 5 ✓

b. 3

c. 23

d. 11

Complete the square: $f(x) = 2(x^2 - 6x) + 23 = 2((x - 3)^2 - 9) + 23 = 2(x - 3)^2 - 18 + 23 = 2(x - 3)^2 + 5$. The vertex is $(3, 5)$. Because $a = 2 > 0$, the minimum value is $k = 5$.

35 [Intermediate] • Target: Polynomial division and the remainder theorem

If $x = 3$ is a solution to the quadratic equation $2x^2 + kx - 15 = 0$, what is the value of the constant k?

a. -1 ✓

b. 1

c. -3

d. 3

Substitute $x = 3$ into the equation: $2(3)^2 + k(3) - 15 = 0 \Rightarrow 2(9) + 3k - 15 = 0 \Rightarrow 18 - 15 + 3k = 0 \Rightarrow 3 + 3k = 0 \Rightarrow 3k = -3 \Rightarrow k = -1$.

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36 [Intermediate] • Target: Quadratic word problems

Consider the quadratic function $f(x) = x^2 - 6x + 9$. What is the value of $f(3)$?

a. 0 ✓

b. 3

c. 9

d. 18

To find the value of $f(3)$, we substitute $x = 3$ into the function $f(x) = x^2 - 6x + 9$. $f(3) = (3)^2 - 6(3) + 9$ $f(3) = 9 - 18 + 9$ $f(3) = 0$ Therefore, the value of $f(3)$ is 0.

37 [Intermediate] • Target: Sum and product of roots

Simplify $\frac{3+4i}{1-2i}$ into Cartesian form $a + bi$.

a. $-1 + 2i$ ✓

b. $1 + 2i$

c. $-1 - 2i$

d. $2 + i$

Multiply numerator and denominator by $1 + 2i$: $\frac{(3+4i)(1+2i)}{1-(-4)} = \frac{3+6i+4i-8}{5} = \frac{-5+10i}{5} = -1 + 2i$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: A well-designed word problem: maximizing weekly profit • Cross-Lesson: 6.01 Geometry

What is the product of the solutions of $x^3 - 4x^2 - 7x + 10 = 0$?

a. **-10** ✓

b. -12

c. 14

d. 5

By Vieta\x27s formulas, for a polynomial $ax^3 + bx^2 + cx + d = 0$, the product of the roots is $-\frac{d}{a} = -\frac{10}{1} = -10$.

39 [Challenging] • Target: Factoring quadratics • Cross-Lesson: 7.01 Trigonometry & Additional Topics

What is the sum of the solutions to $|x^2 - 7| = 9$?

a. 0 ✓

b. 8

c. 4

d. 16

$x^2 - 7 = 9 \Rightarrow x^2 = 16 \Rightarrow x = \pm 4$. $x^2 - 7 = -9 \Rightarrow x^2 = -2$ (no real roots). Sum is $4 + (-4) = 0$.

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40 [Challenging] • Target: Vertex form and completing the square • Cross-Lesson: 8.01 Exponentials & Logarithms

What is the sum of the solutions of $(a-1)(a^2+8a+15)=0$?

a. 8

b. 7

c. **-7** ✓

d. -8

Solutions: $a=1$ and roots of $a^2+8a+15=(a+3)(a+5)=0 \Rightarrow a=-3,\ -5$. Sum $=1-3-5=-7$.

41 [Challenging] • Target: The quadratic formula and discriminant • Cross-Lesson: 1.01 Heart of Algebra

What is the vertex of the parabola defined by $y=2x^2-12x+11$?

a. $(3,\ -7)$ ✓

b. $(3, 11)$

c. $(-3,\ -7)$

d. $(6,\ -7)$

$h=-b/(2a)=12/4=3$. $k=2(9)-12(3)+11=18-36+11=-7$. Vertex is $(3,\ -7)$.

42 [Challenging] • Target: Graphing parabolas • Cross-Lesson: 3.01 Functions & Graphs

If α and β are the roots of $x^2-5x+3=0$, what is the value of $\frac{1}{\alpha}+\frac{1}{\beta}$?

a. $\frac{5}{3}$ ✓

b. $\frac{3}{5}$

c. $-\frac{5}{3}$

d. 5

$\frac{1}{\alpha}+\frac{1}{\beta}=\frac{\alpha+\beta}{\alpha\beta}=\frac{5}{3}$.

43 [Challenging] • Target: Polynomial multiplication and special products • Cross-Lesson: 4.01 Ratios, Rates & Data

If the polynomial x^4+ax^2+b is divisible by x^2+3x+2, what is $a+b$?

a. -1 ✓

b. 1

c. 5

d. -5

Factors are $x=-1$ and $x=-2$. At -1: $1+a+b=0 \Rightarrow a+b=-1$.

44 [Challenging] • Target: Polynomial division and the remainder theorem • Cross-Lesson: 5.01 Statistics & Probability

What is the minimum value attained by the quadratic function $f(x)=3x^2-12x+19$?

a. 7 ✓

b. 19

c. 2

d. 3

Vertex $x=-(-12)/(6)=2$. Minimum value $=f(2)=3(4)-24+19=7$.

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45 [Challenging] • Target: Quadratic word problems • Cross-Lesson: 6.01 Geometry

The graph of the quadratic function $g(x) = ax^2 + bx + c$ is shown below. The parabola passes through the points $(-2, 10)$ and $(4, 10)$, and its vertex is at $(1, -8)$. Determine the value of a.

a. 1

b. **2** ✓

c. -1

d. -2

The vertex of a parabola $g(x) = ax^2 + bx + c$ is given by $(h, k) = (-\frac{b}{2a}, g(-\frac{b}{2a}))$. From the graph, the vertex is $(1, -8)$, so $h = 1$ and $k = -8$. Since $h = 1$, we have $-\frac{b}{2a} = 1$, which implies $b = -2a$. The function can be written in vertex form as $g(x) = a(x - h)^2 + k$. Substituting the vertex coordinates $(1, -8)$, we get: $g(x) = a(x - 1)^2 - 8$. The parabola passes through the point $(-2, 10)$. Substitute these coordinates into the vertex form equation: $10 = a(-2 - 1)^2 - 8$.

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